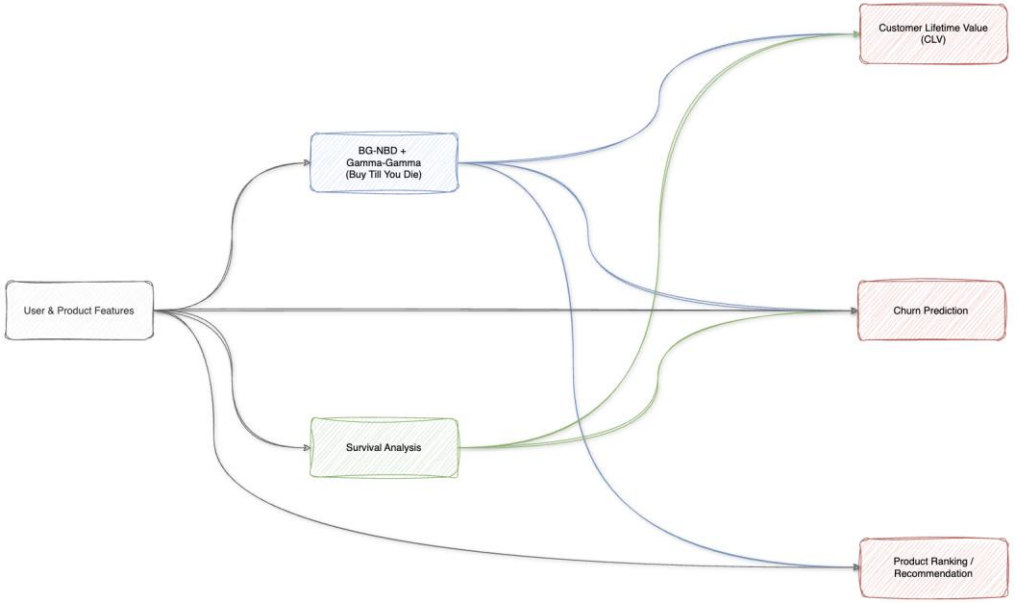
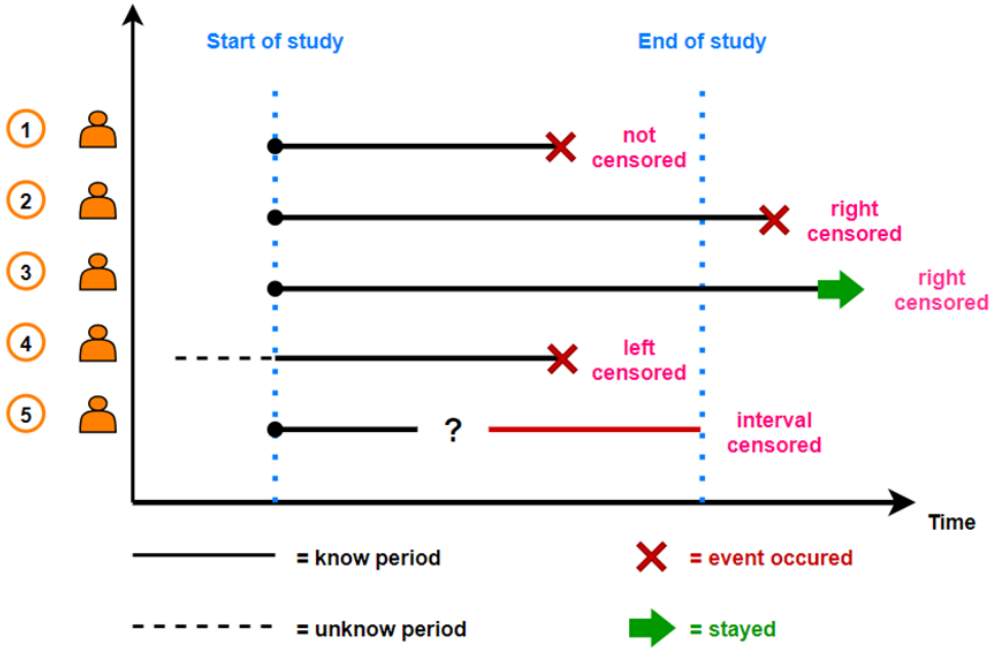
Survival Analysis

Lê Thiện Hoà

Why Survival Analysis ?

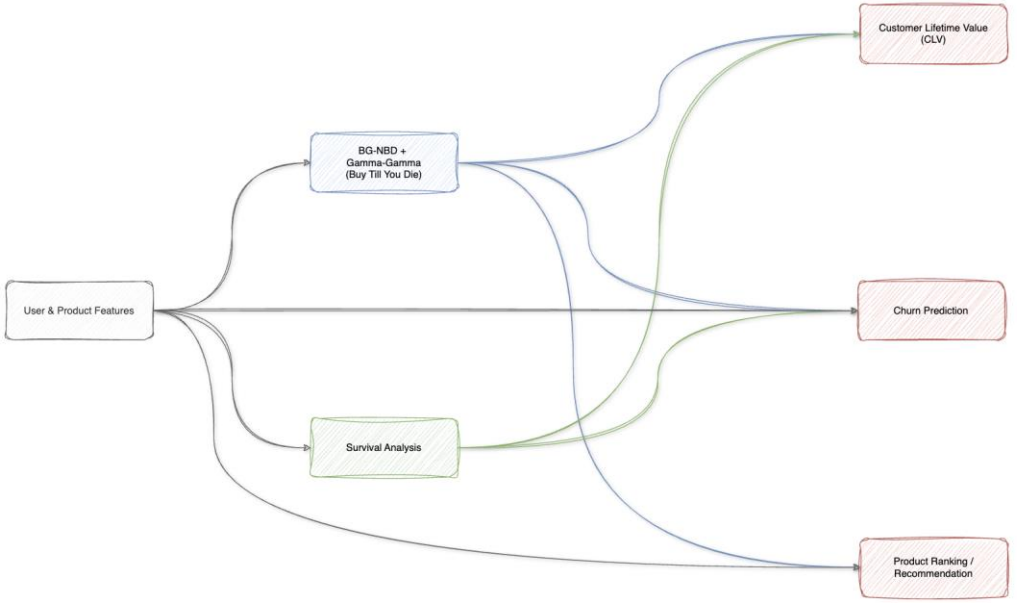
- Many business questions involve **time-to-event** (churn, repayment, upgrade, fraud).
- Traditional models (classification / regression) ignore **censoring**.
- Survival Analysis provides:
  - **Right-censoring** (customer still active today)
  - **Probability of churn at each future time point** (individual survival curve)
  - **Expected CLV** when combined with revenue model

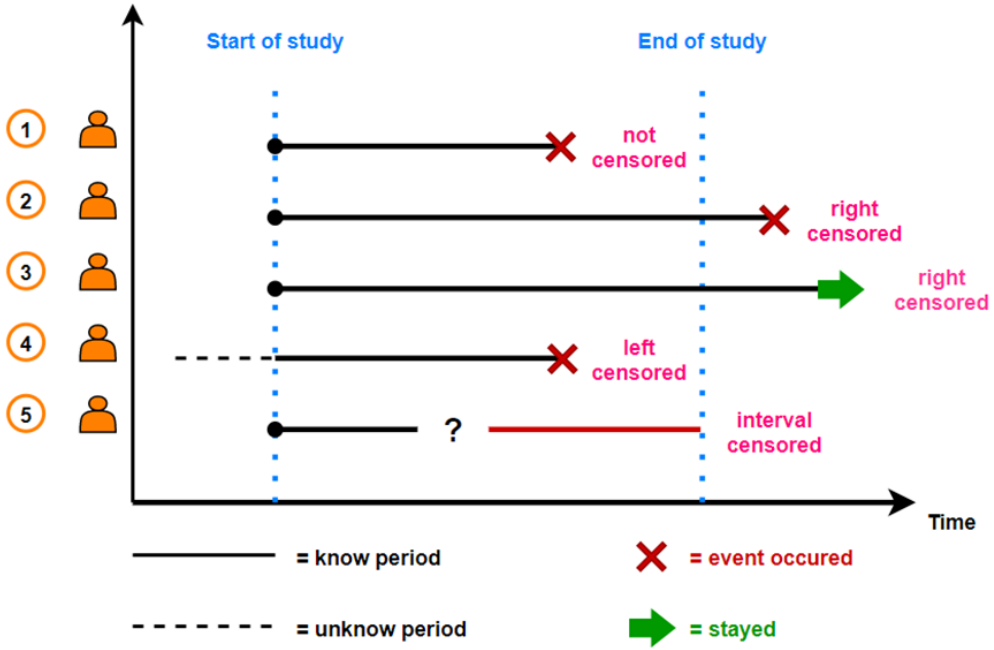




Key Concepts

- **Event**: churn, conversion, default, cancellation.
- **Time-to-event:** days until churn or still active.
- **Censoring:** event not observed yet (still using product).
- **Hazard:** instantaneous risk of churn at time t.
- **Survival probability:** chance a user is still active after t days.





Censoring

Right Censoring

Right censoring happens when study ends but no event is observed. The patient did regular follow-ups in the study period. The event might have occurred after the study, but actual time could not be noted and hence it is censored. The below diagram explains the situation.

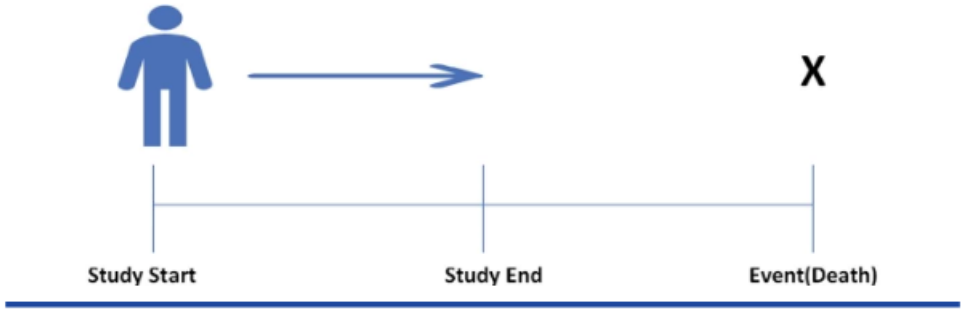


Figure 1.1 Event sequence for right censoring.

Left Censoring

Left censoring happens when event has already occurred before the start of the study. The following diagram depicts the situation.

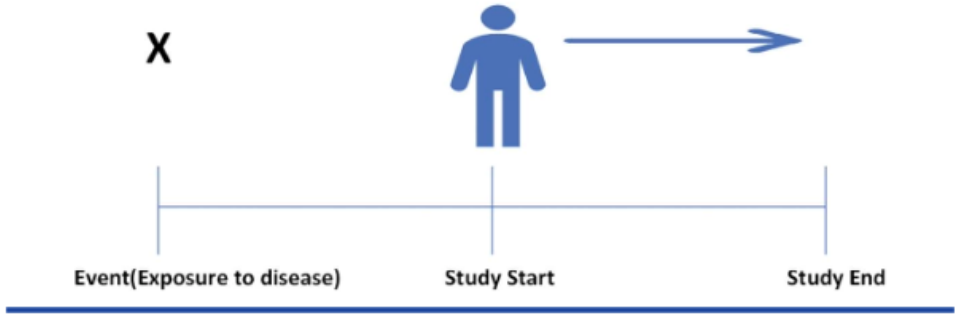


Figure 1.2 Event sequence for left censoring.

Interval Censoring

Interval censoring happens when event occurs within the study period in between two possible time limits, and as usual actual time could not be noted. The below diagram explains the situation.

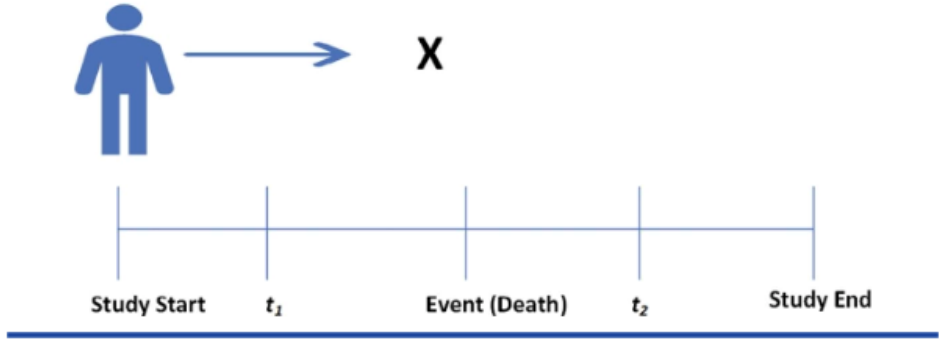
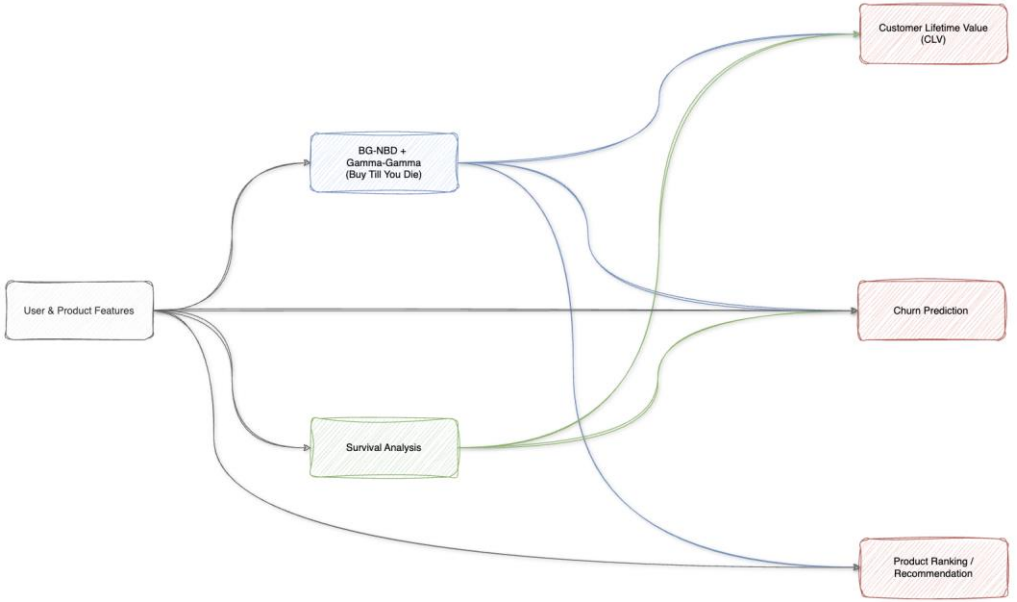
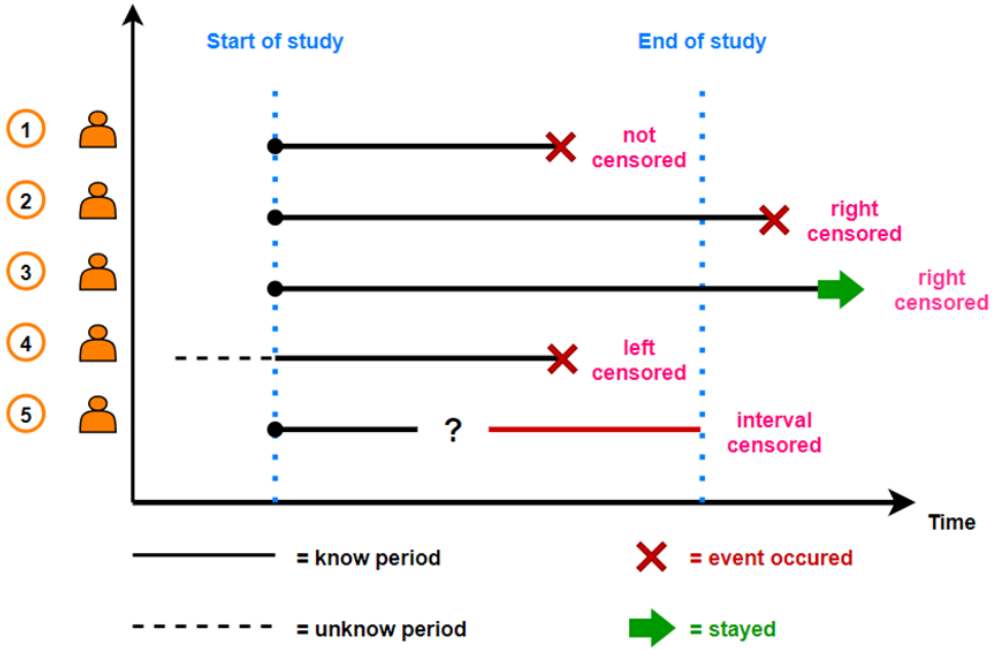


Figure 1.3 Event sequence for interval censoring.

Why Not Linear Regression for Churn Time?

- Cannot handle **censored users** → biased estimates.
- Linear Regression can predict **negative time**.
- Assumes **constant variance** and **linear trend** with **normal distribution assumption for residuals** - **unrealistic**.
- Survival models **explicitly model risk over time** → more reliable churn-time estimation.





Core Modeling Approaches

1. **Parametric Models**
    - Exponential, Weibull, Log-Logistic
2. **Non-parametric Models**
    - Kaplan Meier
3. **Semi-parametric Models**
    - Cox Proportional Hazards (CoxPH)
        - Non Time-varying features
        - Time-varying features / covariates
        - Time-varying coefficients
    - DeepSurv
4. **Multi-Task Logistic Regression (MLTR)**
    - Linear MLTR
    - Neural MLTR
5. **Machine Learning Survival Models**
    - Random Survival Forests, Gradient Boosting Survival, Conditional Survival Forests
    - DeepHit, Deep Cox (Cox-Time)

0. Prerequisites – T, Survival Time

T is a positive random variable representing the waiting time until an event occurs. Its probability density function (p.d.f.) is $f(t)$ and cumulative distribution function (c.d.f.) is given by

$$F(t) = \Pr[T < t] = \int_{-\infty}^{t} f(u)du$$

- **Example:** Let's take the example of credit risk and assume that the event of interest is `fully repaying a loan`. We can now analyze the cumulative distribution function of two distinct borrowers through time.

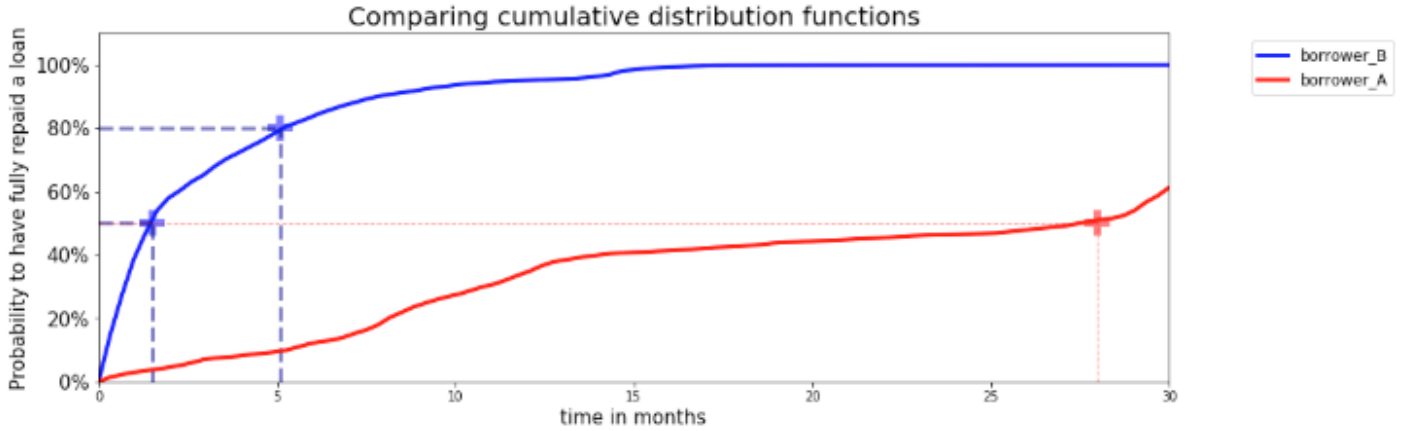


Figure 1 - Comparing cumulative distribution functions

Here, we can see that the probability that Borrower B has fully repaid his/her loan reaches 50% or 80% much faster than Borrower A's. This indicates that Borrower B is potentially less risky than Borrower A.

0. Prerequisites – $S(t)$, Survival Function

$S(t)$ is the probability that the event of interest has not occurred by some time t

$$S(t) = 1 - F(t) = \Pr[T \geq t]$$

- **Example:** Here, we will consider the example of churn modeling, assume that the event of interest is `stopping the SaaS subscription` and analyze the survival function of three distinct customers through time.

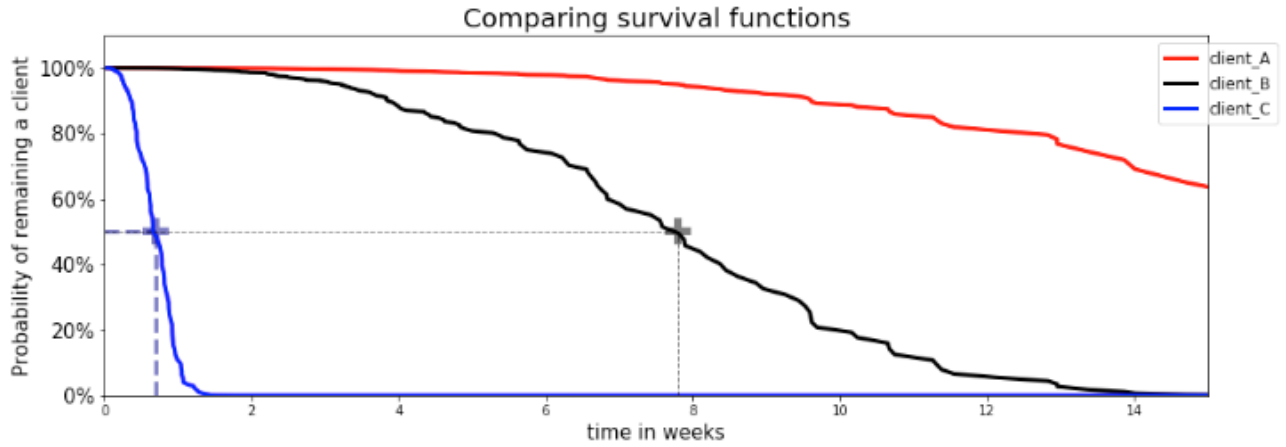


Figure 2 - Comparing survival functions

Here, we can see that the probability of remaining a customer reaches 50% much faster for Client C than Client B. On the other hand, Client A's probability doesn't even go below 60% from week 0 to week 15 of the analysis. In a nutshell,

- Client C is very likely to churn within the first 2 weeks
- Client B is likely to churn within the next 15 weeks
- Client A is very likely to remain a customer within the next 15 weeks.

0. Prerequisites – $h(t)$, Hazard Function

Hazard Function $h(t)$: The instantaneous conditional event rate, defined as the probability of an event occurring in the next small interval of time (from t to $t + \Delta t$) given that the individual has survived until time t, divided by the interval length Δt as Δt approaches zero.

$$h(t) = \lim_{\Delta t \to 0} \frac{P(t \le T < t + \Delta t | T \ge t)}{\Delta t}$$

Using the rules of conditional probability:

$$P(t \le T < t + \Delta t | T \ge t) = \frac{P(t \le T < t + \Delta t \cap T \ge t)}{P(T \ge t)}$$

The numerator is approximately $f(t)\Delta t$ (for a small interval), and the denominator is $S(t)$

So, the definition of the hazard function can be written as:

$$h(t) = \frac{f(t)}{S(t)}$$

or

$$S(t) = \exp\left(-\int_0^t h(u)\, du\right)$$

- **S(t)**: likelihood of still being alive up to time t
- **H(t)**: likelihood of death at that time t

- **High** Hazard ratio **->** Survival **drops quickly**
- **Small** Hazard ratio -> Survival **decreases slowly**

1. Parametric Models

Assumption: hazard rate follows a **specific distribution**

Exponential

$$\lambda = \alpha e^{\overrightarrow{x_i} \cdot \vec{\omega}}$$
$$h(t, x_i) = \lambda$$
$$S(t, x_i) = e^{-\lambda \cdot t}$$

with α and $\vec{\omega}$ the coefficients to find.

hazard rate constant over time

Weibull

$$\lambda = \alpha e^{\overrightarrow{x_i} \cdot \vec{\omega}}$$
$$h(t, x_i) = \lambda \beta (\lambda t)^{\beta - 1}$$
$$S(t, x_i) = e^{-(\lambda t)^{\beta}}$$

with α, β and $\vec{\omega}$ the coefficients to find.

hazard rate increases / decreases over time

Log-Logistic

$$\lambda = \alpha e^{\overrightarrow{x_i} \cdot \vec{\omega}}$$
$$h(t, x_i) = \frac{\beta \lambda^{\beta} t^{\beta - 1}}{1 + (\lambda t)^{\beta}}$$
$$S(t, x_i) = \frac{1}{1 + (\lambda t)^{\beta}}$$

with α, β and $\vec{\omega}$ the coefficients to find.

hazard rate increases then decreases over time

1. Parametric Models – Exponential Model

Assumption: hazard rate is constant over time (i.e.: churn risk today = tomorrow = day after tomorrow,….)

- **High Hazard rate:** churn soon
- **Low Hazard rate**: live long

Exponential

Hazard function: $h(t) = \lambda$ (where λ is the constant hazard rate or event rate)

Survival function: $S(t) = e^{-\lambda t}$

Probability Density Function (PDF): $f(t) = h(t)S(t) = \lambda e^{-\lambda t}$

Construct Likelihood Function:

For a dataset with n observations, each observation i has an observed time t_i and an event indicator δ_i ($\delta_i = 1$ if an event occurred, $\delta_i = 0$ if censored).

- If an event is observed at time t_i, its contribution to the likelihood is the PDF: $L_i = f(t_i) = \lambda e^{-\lambda t_i}$.
- If an observation is censored at time t_i, we know the event time is greater than t_i. Its contribution is the survival function: $L_i = S(t_i) = e^{-\lambda t_i}$.

The overall likelihood function $L(\lambda)$ is the product of the individual contributions:

$$L(\lambda) = \prod_{i=1}^{n} [f(t_i)]^{\delta_i} [S(t_i)]^{1-\delta_i}$$

1. Parametric Models – Exponential Model

Substituting the exponential functions:

$$L(\lambda) = \prod_{i=1}^{n} [\lambda e^{-\lambda t_i}]^{\delta_i} [e^{-\lambda t_i}]^{1-\delta_i}$$

$$L(\lambda) = \prod_{i=1}^{n} \lambda^{\delta_i} e^{-\lambda t_i \delta_i} e^{-\lambda t_i (1-\delta_i)}$$

$$L(\lambda) = \prod_{i=1}^{n} \lambda^{\delta_i} e^{-\lambda t_i \delta_i - \lambda t_i + \lambda t_i \delta_i}$$

$$L(\lambda) = \prod_{i=1}^{n} \lambda^{\delta_i} e^{-\lambda t_i}$$

It is mathematically easier to work with the natural logarithm of the likelihood function (log-likelihood), denoted as $l(\lambda)$. Maximizing the log-likelihood is equivalent to maximizing the likelihood

$$l(\lambda) = \ln(L(\lambda)) = \ln\left(\prod_{i=1}^{n} \lambda^{\delta_i} e^{-\lambda t_i}\right)$$

Using log properties: $\ln(AB) = \ln(A) + \ln(B)$ and $\ln(A^b) = b\ln(A)$:

$$l(\lambda) = \sum_{i=1}^{n} \ln(\lambda^{\delta_i}) + \sum_{i=1}^{n} \ln(e^{-\lambda t_i})$$

$$l(\lambda) = \sum_{i=1}^{n} \delta_i \ln(\lambda) + \sum_{i=1}^{n} (-\lambda t_i)$$

$$l(\lambda) = (\sum_{i=1}^{n} \delta_i) \ln(\lambda) - \lambda (\sum_{i=1}^{n} t_i)$$

Let $d = \sum_{i=1}^{n} \delta_i$ be the total number of observed events, and $T = \sum_{i=1}^{n} t_i$ be the total person-time observed.

$$l(\lambda) = d\ln(\lambda) - \lambda T$$

1. Parametric Models – Exponential Model

Maximize the Log-Likelihood Function

To find the maximum likelihood estimator (MLE) for λ, denoted as $\hat{\lambda}$, we take the first derivative of the log-likelihood function with respect to λ (the score function), set it to zero, and solve for λ.

$$\frac{d}{d\lambda} l(\lambda) = \frac{d}{d\lambda} (d \ln(\lambda) - \lambda T)$$

$$\frac{d}{d\lambda} l(\lambda) = \frac{d}{\lambda} - T$$

Set the derivative to zero:

$$\frac{d}{\hat{\lambda}} - T = 0$$

$$\frac{d}{\hat{\lambda}} = T$$

$$\hat{\lambda} = \frac{d}{T}$$

1. Parametric Models – Exponential Model

Model with Covariates

The hazard function for an individual i with covariate vector X_i is typically modeled using a proportional hazards form to ensure the hazard remains positive:

$$h_i(t|X_i) = \lambda_0 \exp(X_i\beta)$$

Where λ_0 is the constant baseline hazard (the hazard when all covariates are zero), and β is a vector of coefficients. For the exponential model, the baseline hazard λ_0 is a single positive constant (the time-independent rate parameter).

The survival function and PDF for individual i become:

- **Survival Function:** $S_i(t_i|X_i) = \exp(-\int_0^{t_i} h_i(u|X_i)du) = \exp(-\lambda_0 t_i \exp(X_i\beta))$
- **PDF:** $f_i(t_i|X_i) = h_i(t_i|X_i)S_i(t_i|X_i) = \lambda_0 \exp(X_i\beta)\exp(-\lambda_0 t_i \exp(X_i\beta))$

1. Parametric Models – Exponential Model

Model with Covariates

Log-Likelihood Derivation with Covariates

We follow the same general steps as before:

Step 1: Formulate the Likelihood Function

The likelihood is the product of contributions from observed events ($\delta_i = 1$) and censored observations ($\delta_i = 0$):

$$L(\lambda_0, \beta|\text{Data}) = \prod_{i=1}^{n} [f_i(t_i|X_i)]^{\delta_i} [S_i(t_i|X_i)]^{1-\delta_i}$$

Substituting the exponential functions:

$$L = \prod_{i=1}^{n} [\lambda_0 \exp(X_i\beta) \exp(-\lambda_0 t_i \exp(X_i\beta))]^{\delta_i} [\exp(-\lambda_0 t_i \exp(X_i\beta))]^{1-\delta_i}$$

$$L = \prod_{i=1}^{n} (\lambda_0 \exp(X_i\beta))^{\delta_i} \exp(-\lambda_0 t_i \exp(X_i\beta))$$

Step 2: Formulate the Log-Likelihood Function

Take the natural logarithm of L:

$$l(\lambda_0, \beta) = \sum_{i=1}^{n} [\delta_i \ln(\lambda_0) + \delta_i X_i\beta - \lambda_0 t_i \exp(X_i\beta)]$$

$$l(\lambda_0, \beta) = (\sum_{i=1}^{n} \delta_i) \ln(\lambda_0) + (\sum_{i=1}^{n} \delta_i X_i\beta) - \lambda_0 (\sum_{i=1}^{n} t_i \exp(X_i\beta))$$

Let $d = \sum \delta_i$ be the total number of events.

$$l(\lambda_0, \beta) = d \ln(\lambda_0) + \sum_{i=1}^{n} \delta_i X_i\beta - \lambda_0 \sum_{i=1}^{n} t_i \exp(X_i\beta)$$

1. Parametric Models – Exponential Model

Model with Covariates

Step 3: Maximization (Numerical Optimization Required)

To find the MLEs for λ_0 and β, we take the partial derivatives of the log-likelihood with respect to each parameter (λ_0 and each element of β), set them to zero, and solve.

- **Partial derivative with respect to λ_0:**

  $$\frac{\partial l}{\partial \lambda_0} = \frac{d}{\lambda_0} - \sum_{i=1}^{n} t_i \exp(X_i \beta)$$

  Setting this to zero and solving for $\hat{\lambda}_0$ gives a *conditional* closed-form solution:

  $$\hat{\lambda}_0 = \frac{d}{\sum_{i=1}^{n} t_i \exp(X_i \beta)}$$

  This equation shows that the estimate for the baseline hazard depends on the estimates of the β coefficients.

- **Partial derivative with respect to β_j** (for a specific covariate j):

  $$\frac{\partial l}{\partial \beta_j} = \sum_{i=1}^{n} \delta_i X_{ij} - \lambda_0 \sum_{i=1}^{n} t_i X_{ij} \exp(X_i \beta)$$

  Setting this to zero:

  $$\sum_{i=1}^{n} \delta_i X_{ij} = \lambda_0 \sum_{i=1}^{n} t_i X_{ij} \exp(X_i \beta)$$

The set of equations for the β coefficients is non-linear and cannot be solved algebraically for a simple closed-form solution. The equations are inter-dependent with λ_0 and the other βs.

Therefore, for the exponential model with covariates, you must use **iterative numerical methods** to find the values of $\hat{\beta}$ and $\hat{\lambda}_0$ that satisfy the system of score equations (first derivatives set to zero).

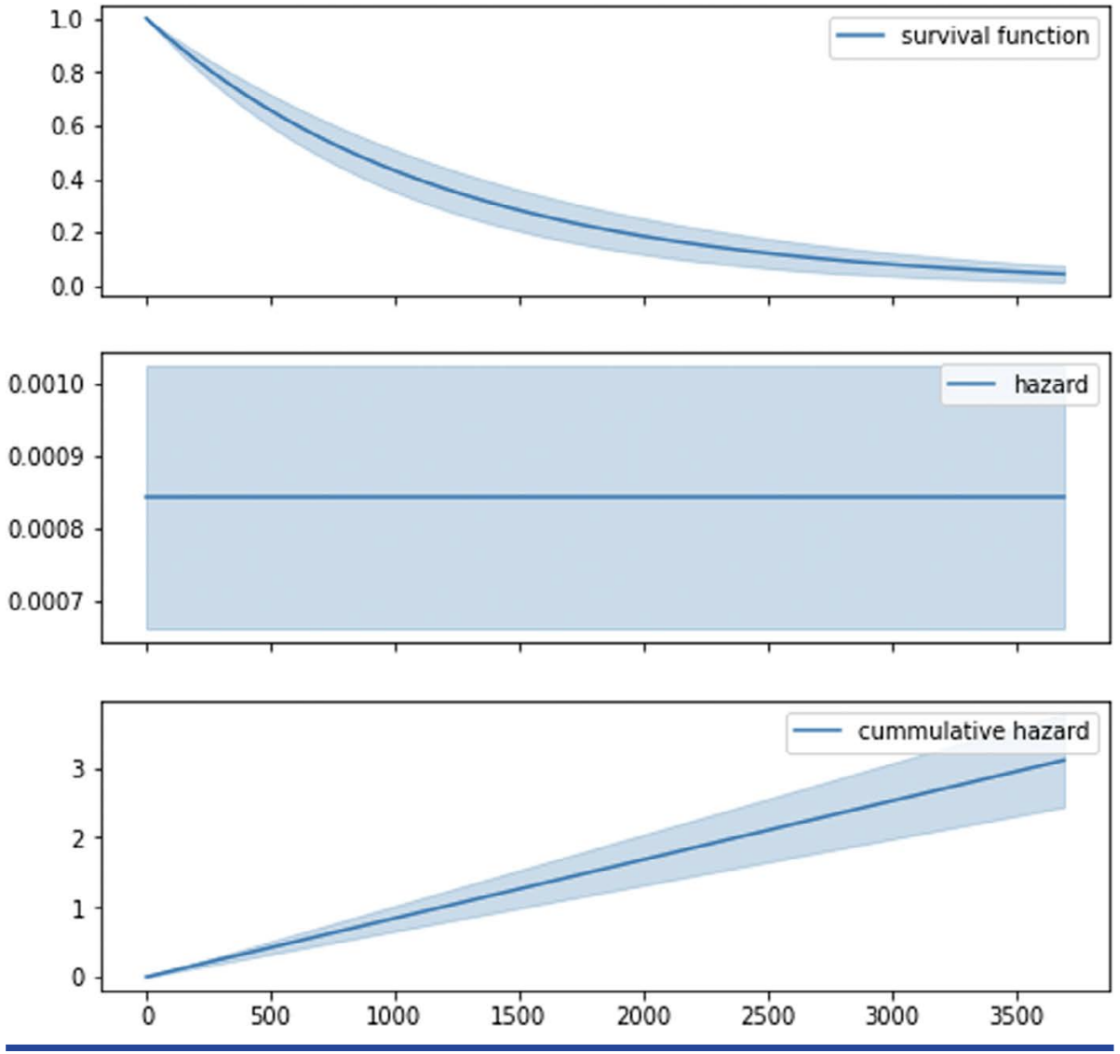


Figure 3.6 **Survival, hazard, cumulative hazard plot for *exponential*.**

2. Non-Parametric Models : **Kaplan-Meier**

Assumption: do not make any assumption about survival time distribution (unlike the parametric model)

- Only used for estimating past data, not for prediction

The Kaplan-Meier estimator is based on the product limit method. The core idea is that the probability of surviving past a certain time point t is the product of the conditional probabilities of surviving each preceding time interval.

The estimated survival function $\hat{S}(t)$ is calculated as:

$$\hat{S}(t) = \prod_{t_i \le t} (1 - \hat{h}(t_i)) = \prod_{t_i \le t} \left(1 - \frac{d_i}{n_i}\right)$$

Where, at each unique time point t_i:

- n_i = the number of individuals alive and at risk *just before* time t_i (the "risk set").
- d_i = the number of events (e.g., deaths, failures) observed *at* time t_i.
- $\frac{d_i}{n_i}$ is the estimated hazard ($\hat{h}(t_i)$) at that specific time point.
- $1 - \frac{d_i}{n_i}$ is the conditional probability of surviving past time t_i, given survival up to that point.

The model assumes that censoring is non-informative (unrelated to prognosis) and that events occur at the precise times they are observed.

Table 4.1 Step-Wise Computation of KM Estimator

t_i	d_i	n_i	d_i/n_i	$S_{KM}(t)$
0.5	1	184	1/184	$\left(1-\frac{1}{184}\right)$
1.0	2	183	2/183	$\left(1-\frac{1}{184}\right)\times\left(1-\frac{2}{183}\right)$
2.0	1	181	1/181	$\left(1-\frac{1}{184}\right)\times\left(1-\frac{2}{183}\right)\times\left(1-\frac{1}{181}\right)$

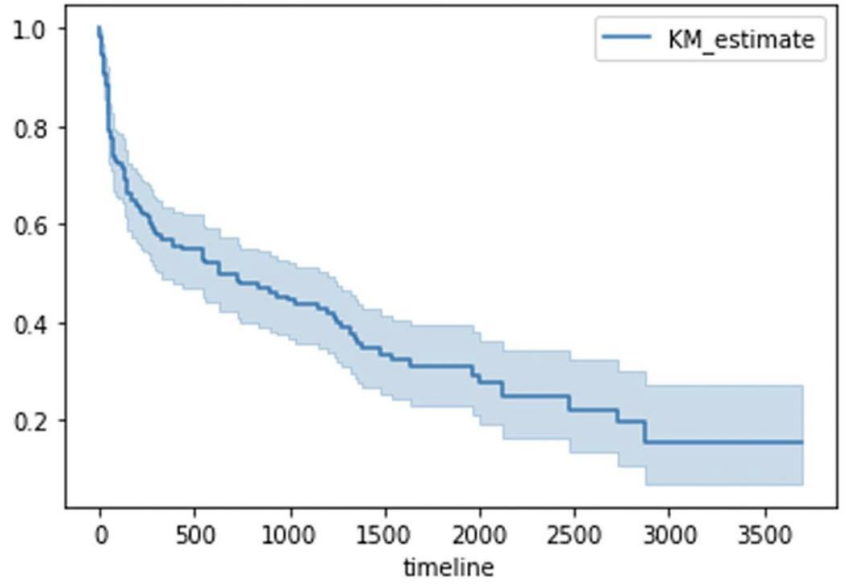


Figure 4.2 Survival function plot of fitted KM model.

Due to its discrete nature, KM estimate curves always produce step-wise looks.

3. Semi-Parametric Models - **Cox Proportional Hazards (CoxPH)**

Assumption:

1. The hazard ratio between any two individuals is **constant over time**
   - Covariates have a **multiplicative** effect on the hazard
   - The effect **does not change over time** (i.e., ***no time-varying effect***)
2. **Parametric**: **Linearity of covariates** in the log-hazard
   - **Non-Linear Relationship ->** use **DeepSurv**
3. **Non-Parametric**: **baseline hazard** is **unspecified**, but **shared**
   - All individuals share the same baseline hazard, modified by covariates through $\exp(X\beta)$
   - **No distribution assumption** for baseline hazard. No need to estimate function $h_0(t)$
   - No need to know $h_0(t)$ to estimat β . We will **estimate baseline hazard at the final step** via **Breslow / Nelson-Aalen** estimator

The model focuses on the **hazard function**, $h(t|X)$, which represents the instantaneous risk of an event occurring at time t, given that the subject has survived up to that time. The mathematical form of the model is:

$$h(t|X) = h_0(t)\exp(\beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p) = h_0(t)\exp(X\beta)$$

where:

- $h(t|X)$ is the hazard function for a subject with covariate values $X = (x_1, \ldots, x_p)$.
- $h_0(t)$ is the **baseline hazard function**, which is the hazard for a subject when all covariates are zero. It is a non-parametric component and its shape is not assumed to follow any specific distribution (making the model semi-parametric).
- $\beta = (\beta_1, \ldots, \beta_p)$ is the vector of **regression coefficients** that quantify the effect of the covariates.

The model is most commonly interpreted in terms of the **hazard ratio (HR)**. The ratio of hazards for two individuals, say with covariate sets X_i and X_j, is:

$$HR = \frac{h(t|X_i)}{h(t|X_j)} = \frac{h_0(t)\exp(X_i\beta)}{h_0(t)\exp(X_j\beta)} = \exp((X_i - X_j)\beta)$$

The baseline hazard $h_0(t)$ cancels out, showing that the hazard ratio is constant over time t and dependent only on the covariates and their coefficients.

3. Semi-Parametric Models - **Cox Proportional Hazards (CoxPH)**

Maximum Likelihood

1. The Full Likelihood Structure

The full likelihood function for survival data is constructed using the probability density functions ($f(t)$) for uncensored observations and the survival functions ($S(t)$) for censored observations.

The full likelihood $L_{full}(\beta, \lambda_0(\cdot))$ for a dataset with n subjects is a product over all subjects i:

$$L_{full}(\beta, \lambda_0(\cdot)) = \prod_{i=1}^{n} \left[f_i(T_i)\right]^{\delta_i} \left[S_i(T_i)\right]^{1-\delta_i}$$

- T_i is the observation time (event or censoring time).
- δ_i is the event indicator ($\delta_i = 1$ if an event occurred, $\delta_i = 0$ if censored).

2. Expressing the Full Likelihood in Terms of Hazard

We use the fundamental relationships between survival functions, density functions, and the cumulative hazard function ($\Lambda(t)$):

- $S(t) = \exp(-\Lambda(t))$
- $f(t) = \lambda(t)S(t)$

Substituting the Cox PH model $\lambda_i(t) = \lambda_0(t)e^{X_i\beta}$ into these formulas gives us the full likelihood entirely in terms of the baseline hazard and coefficients:

$$L_{full}(\beta, \lambda_0(\cdot)) = \prod_{i=1}^{n} \left[\lambda_0(T_i)e^{X_i\beta}S_i(T_i)\right]^{\delta_i} \left[S_i(T_i)\right]^{1-\delta_i}$$

This simplifies to:

$$L_{full}(\beta, \lambda_0(\cdot)) = \left(\prod_{i:\delta_i=1} \lambda_0(T_i)\right)\left(\prod_{i=1}^{n} e^{\delta_i X_i\beta}\right)\left(\prod_{i=1}^{n} S_i(T_i)\right)$$

The Problem with Maximization

The term $\lambda_0(T_i)$ is entirely unspecified. The maximum likelihood approach typically requires you to take derivatives with respect to all parameters (β and $\lambda_0(t)$) and set them to zero.

3. Semi-Parametric Models - **Cox Proportional Hazards (CoxPH)**

Maximum ***Partial*** Likelihood

Steps for Derivation:

1. **Order Event Times**: Assume there are n subjects and m distinct observed event (e.g., death) times $t_{(1)} < t_{(2)} < \cdots < t_{(m)}$. For simplicity, assume no tied event times in the basic derivation.
2. **Define Risk Set**: At each event time $t_{(i)}$, define the **risk set** $R(t_{(i)})$ as the set of all subjects who are alive and uncensored just before time $t_{(i)}$.
3. **Conditional Probability**: For each event time $t_{(i)}$, consider the probability that the specific subject who failed (let's call them subject l) is the one who fails, out of all subjects currently in the risk set $R(t_{(i)})$.
    1. The probability that subject l has an event at time $t_{(i)}$, given that *one* event occurs in the risk set $R(t_{(i)})$ at that time, is:

    $$P(\text{Subject } l \text{ fails} | \text{one event in } R(t_{(i)})) = \frac{h(t_{(i)}|X_l)}{\sum_{j \in R(t_{(i)})} h(t_{(i)}|X_j)}$$

    2. Substituting the Cox model formula:

    $$= \frac{h_0(t_{(i)}) \exp(X_l\beta)}{\sum_{j \in R(t_{(i)})} h_0(t_{(i)}) \exp(X_j\beta)} = \frac{\exp(X_l\beta)}{\sum_{j \in R(t_{(i)})} \exp(X_j\beta)}$$

    3. The baseline hazard $h_0(t_{(i)})$ cancels out in the ratio, which is the crucial step that removes the need to estimate it.
4. **Form Partial Likelihood**: The partial likelihood is the product of these conditional probabilities over all observed event times.

$$L_p(\beta) = \prod_{i=1}^{m} \left[\frac{\exp(X_l\beta)}{\sum_{j \in R(t_{(i)})} \exp(X_j\beta)} \right]$$

The **log-partial likelihood** $L(\beta) = \ln(L_p(\beta))$ simplifies to a sum of logs:

$$L(\beta) = \sum_{i=1}^{m} \left(X_l\beta - \ln \left(\sum_{j \in R(t_{(i)})} \exp(X_j\beta) \right) \right)$$

where X_l is the covariate vector for the subject who experienced the event at time $t_{(i)}$

Log-Sum-Exp: sum exponential can not invert log to split variable

➔ **No closed-form solution**

➔ Need to use **Iterative Method**

-> Instead of modeling the **Full Likelihood**, we **only consider the probability** that the individual who actually **died at that time** among **all individuals still at risk** just before that event time

3. Semi-Parametric Models - **Cox Proportional Hazards (CoxPH)**

Using the Newton-Raphson Method

The Newton-Raphson method is an iterative numerical algorithm used to find the roots of a function (in this case, the root of the score function $U(\beta) = 0$, which corresponds to the maximum of the log-likelihood $L(\beta)$).

The Algorithm Steps:

1. **Start with an Initial Guess**: Begin with an initial estimate $\beta^{(0)}$, often a vector of zeros.
2. **Iterative Update**: Update the estimate using the following formula:
   $\beta^{(k+1)} = \beta^{(k)} - H(\beta^{(k)})^{-1} U(\beta^{(k)})$
   where $H(\beta^{(k)})^{-1}$ is the inverse of the Hessian matrix evaluated at the current estimate $\beta^{(k)}$.
3. **Repeat**: Repeat the iteration until the estimates converge, meaning the change between $\beta^{(k+1)}$ and $\beta^{(k)}$ is sufficiently small. The final value is the maximum partial likelihood estimate $\hat{\beta}$.

Derivation by β to get the Score Vector and Hessian Matrix

We use vector calculus to find the derivatives.

1. Score Vector (First Derivative)

The first derivative of the log-partial likelihood with respect to the vector β is known as the **score vector** ($U(\beta)$). Setting this to zero provides the estimating equations for $\hat{\beta}$.

$$U(\beta) = \frac{\partial L(\beta)}{\partial \beta} = \sum_{i=1}^{m} \left(X_l - \frac{\sum_{j \in R(t_{(i)})} X_j \exp(X_j \beta)}{\sum_{j \in R(t_{(i)})} \exp(X_j \beta)} \right) = \sum_{i=1}^{m} (X_l - \bar{X}(\beta))$$

Here, $\bar{X}(\beta)$ represents a weighted average of the covariate vectors of subjects in the risk set $R(t_{(i)})$, weighted by their current hazard contribution.

2. Hessian Matrix (Second Derivative)

The **Hessian matrix** ($H(\beta)$) is the matrix of second partial derivatives, which tells us about the curvature of the log-likelihood function. It is a $p \times p$ matrix (where p is the number of covariates).

$$H(\beta) = \frac{\partial^2 L(\beta)}{\partial \beta \partial \beta^T} = -\sum_{i=1}^{m} \left(\frac{\sum_{j \in R(t_{(i)})} X_j X_j^T \exp(X_j \beta)}{\sum_{j \in R(t_{(i)})} \exp(X_j \beta)} - \bar{X}(\beta) \bar{X}(\beta)^T \right)$$

$$H(\beta) = -\sum_{i=1}^{m} Var(\beta | R(t_{(i)}))$$

The Hessian is a negative semi-definite matrix, representing the concavity of the log-likelihood function. This concavity ensures a unique global maximum exists that the optimization algorithm can find.

3. Semi-Parametric Models - **Cox Proportional Hazards (CoxPH)**

Breslow Method to estimate Baseline Hazard

1. The Segment of the Likelihood at Time t_i

Under the discrete time model, the full likelihood can be structured around each unique event time t_i. The portion of the likelihood function relevant to the specific hazard jump h_i at time t_i is determined by the individuals in the risk set $R(t_i)$ just before that time.

For a specific time t_i:

- d_i individuals experience the event (failure/death).
- Y_i individuals are in the risk set $R(t_i)$ (alive and uncensored).
- $Y_i - d_i$ individuals survive past t_i without an event at that specific instant.

The individual hazard probability for subject j at time t_i is $h_i \times \exp(X_j\hat{\beta})$.
The individual survival probability for subject j at time t_i is $1 - h_i \times \exp(X_j\hat{\beta})$.

The contribution to the likelihood at time t_i from all subjects in the risk set $R(t_i)$ is:

$$L_i(h_i) \propto \left(\prod_{j \in D_i} [h_i \exp(X_j\hat{\beta})] \right) \times \left(\prod_{k \in R(t_i) \setminus D_i} [1 - h_i \exp(X_k\hat{\beta})] \right)$$

This is the product of the probabilities of failure for those who failed (D_i) and the probabilities of survival for those who survived at that instant ($R(t_i) \setminus D_i$).

2. Move to the Log-Likelihood

To maximize this expression, we take the natural logarithm (log-likelihood) for easier differentiation:

$$\ln(L_i(h_i)) = d_i \ln(h_i) + \sum_{j \in D_i} X_j\hat{\beta} + \sum_{k \in R(t_i) \setminus D_i} \ln(1 - h_i \exp(X_k\hat{\beta}))$$

(Note: The $\hat{\beta}$ terms are treated as fixed constants here and don't affect the location of the maximum with respect to h_i).

3. Take the Derivative and Set to Zero

We maximize the log-likelihood with respect to h_i by taking the first derivative and setting it to zero (the score equation):

$$\frac{d \ln(L_i(h_i))}{dh_i} = \frac{d_i}{h_i} - \sum_{k \in R(t_i) \setminus D_i} \frac{\exp(X_k\hat{\beta})}{1 - h_i \exp(X_k\hat{\beta})} = 0$$

3. Semi-Parametric Models - **Cox Proportional Hazards (CoxPH)**

Breslow Method to estimate Baseline Hazard

4. Apply the Key Breslow Approximation (Handling Ties)

The exact solution to the above equation requires complex iterative methods (the Kalbfleisch-Prentice method). Breslow suggested a simplification which works well, especially when the number of ties (d_i) is small relative to the risk set size (Y_i).

The approximation simplifies the denominator term: it assumes that the survival probability within the tiny time window is approximately 1 for everyone in the risk set ($1 - h_i \exp(X_k \hat{\beta}) \approx 1$). This approximation allows the summation to include *all* individuals in the risk set $R(t_i)$, not just those who survived the instant.

The score equation simplifies to:

$$\frac{d_i}{h_i} - \sum_{j \in R(t_i)} \exp(X_j \hat{\beta}) \approx 0$$

(The sum now includes those who failed at t_i as well, which is valid under the approximation).

5. Solve for $\hat{h}_i$

Finally, we rearrange the simplified equation to solve for the maximum likelihood estimate of the instantaneous hazard jump, $\hat{h}_i$:

$$\frac{d_i}{\hat{h}_i} = \sum_{j \in R(t_i)} \exp(X_j \hat{\beta})$$

$$\hat{h}_i = \frac{d_i}{\sum_{j \in R(t_i)} \exp(X_j \hat{\beta})}$$

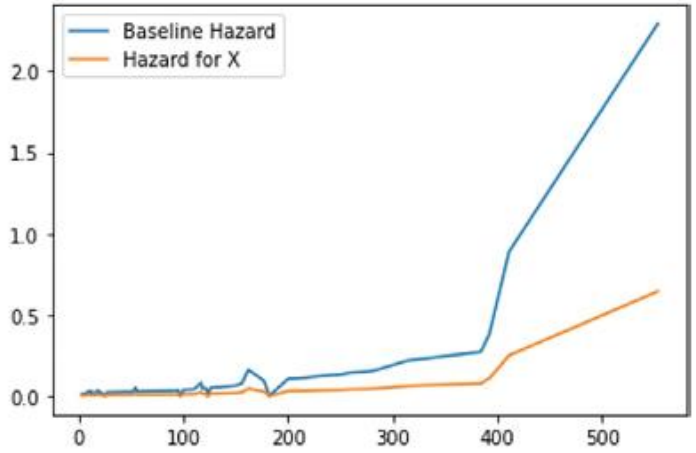


Figure 5.8 **Curves of baseline and original hazard function.**

Evaluation Metrics – **Concordance Index (C-Index)**

The **concordance index** or **C-index** is *a generalization of the area under the ROC curve (AUC)* that can take into account censored data. It represents the global assessment of the model discrimination power: this is the model's ability to correctly provide a reliable ranking of the survival times based on the individual risk scores. It can be computed with the following formula:

$$\text{C-index} = \frac{\sum_{i,j} 1_{T_j<T_i} \cdot 1_{\eta_j>\eta_i} \cdot \delta_j}{\sum_{i,j} 1_{T_j<T_i} \cdot \delta_j}$$

with:

- η_i, the risk score of a unit i
- $1_{T_j<T_i} = 1$ if $T_j < T_i$ else 0
- $1_{\eta_j>\eta_i} = 1$ if $\eta_j > \eta_i$ else 0

Similarly to the AUC, $\text{C-index} = 1$ corresponds to the best model prediction, and $\text{C-index} = 0.5$ represents a random prediction.

Evaluation Metrics – **Concordance Index (C-Index)**

Example Data

Consider five patients with their observed time to event/censoring, event status (1=event, 0=censored), and a model's predicted risk score (higher score means higher predicted risk).

Patient ID	Time (T)	Event Status (Δ)	Predicted Risk (η)
1	5	1	0.8
2	8	1	0.6
3	12	0	0.4
4	15	1	0.3
5	20	0	0.1

Steps to Calculate the C-index

The calculation involves forming all possible pairs of subjects and classifying them.

1. Identify all possible pairs

For 5 patients, there are $5 \times 4/2 = 10$ unique pairs: (1,2), (1,3), (1,4), (1,5), (2,3), (2,4), (2,5), (3,4), (3,5), (4,5).

2. Determine "Comparable Pairs"

A pair is comparable if the subject with the **shorter observation time experienced an event** (was not censored). This ensures an unambiguous time ordering.

Pair (i, j)	T_i, T_j	Δ_i, Δ_j	Shorter Time	Shorter Time Status	Comparable?
(1, 2)	5, 8	1, 1	5 (P1)	1 (Event)	**Yes**
(1, 3)	5, 12	1, 0	5 (P1)	1 (Event)	**Yes**
(1, 4)	5, 15	1, 1	5 (P1)	1 (Event)	**Yes**
(1, 5)	5, 20	1, 0	5 (P1)	1 (Event)	**Yes**
(2, 3)	8, 12	1, 0	8 (P2)	1 (Event)	**Yes**
(2, 4)	8, 15	1, 1	8 (P2)	1 (Event)	**Yes**
(2, 5)	8, 20	1, 0	8 (P2)	1 (Event)	**Yes**
(3, 4)	12, 15	0, 1	12 (P3)	0 (Censored)	**No**
(3, 5)	12, 20	0, 0	12 (P3)	0 (Censored)	**No**
(4, 5)	15, 20	1, 0	15 (P4)	1 (Event)	**Yes**

Total Comparable Pairs = 8

Evaluation Metrics – **Concordance Index (C-Index)**

Example Data

Consider five patients with their observed time to event/censoring, event status (1=event, 0=censored), and a model's predicted risk score (higher score means higher predicted risk).

Patient ID	Time (T)	Event Status (Δ)	Predicted Risk (η)
1	5	1	0.8
2	8	1	0.6
3	12	0	0.4
4	15	1	0.3
5	20	0	0.1

3. Classify Comparable Pairs as Concordant or Discordant

A comparable pair is **concordant** if the subject with the *earlier* event time has a *higher* predicted risk score.

- **Concordant:** Model correctly predicts relative risk.
- **Discordant:** Model incorrectly predicts relative risk.
- **Tied:** If risk scores are tied, the pair is counted as 0.5 concordant and 0.5 discordant (or sometimes omitted, depending on the specific implementation). In this simple example, we assume no ties in risk scores.

Pair	Shorter Time (Early Event)	Longer Time (Later Event)	Risk (Early)	Risk (Later)	Concordant?
(1, 2)	P1 (5 days)	P2 (8 days)	0.8	0.6	**Yes** (0.8 > 0.6)
(1, 3)	P1 (5 days)	P3 (12 days)	0.8	0.4	**Yes** (0.8 > 0.4)
(1, 4)	P1 (5 days)	P4 (15 days)	0.8	0.3	**Yes** (0.8 > 0.3)
(1, 5)	P1 (5 days)	P5 (20 days)	0.8	0.1	**Yes** (0.8 > 0.1)
(2, 3)	P2 (8 days)	P3 (12 days)	0.6	0.4	**Yes** (0.6 > 0.4)
(2, 4)	P2 (8 days)	P4 (15 days)	0.6	0.3	**Yes** (0.6 > 0.3)
(2, 5)	P2 (8 days)	P5 (20 days)	0.6	0.1	**Yes** (0.6 > 0.1)
(4, 5)	P4 (15 days)	P5 (20 days)	0.3	0.1	**Yes** (0.3 > 0.1)

Total Concordant Pairs = 8
Total Discordant Pairs = 0
Total Indeterminate/Tied Pairs = 0

4. Calculate the C-index

The C-index is the ratio of concordant pairs to total comparable pairs.

$$\text{C-index} = \frac{\text{Number of Concordant Pairs}}{\text{Total Number of Comparable Pairs}} = \frac{8}{8} = 1.0$$

In this highly simplified example, the model achieved a perfect C-index of **1.0**, indicating it perfectly ranked every comparable patient pair by their actual risk order.

Evaluation Metrics – **Integrated Brier Score (IBS)**

The **Brier score** is used to evaluate the accuracy of a predicted survival function at a given time t; it represents the *average squared distances between the observed survival status and the predicted survival probability* and is always a number between 0 and 1, with 0 being the best possible value.

Given a dataset of N samples, $\forall i \in [\![1, N]\!], (\vec{x}_i, \delta_i, T_i)$ is the format of a datapoint, and the predicted survival function is $\hat{S}(t, \vec{x}_i), \forall t \in \mathbb{R}^+$:

In the absence of right censoring, the Brier score can be calculated such that:

$$BS(t) = \frac{1}{N}\sum_{i=1}^{N}(1_{T_i>t} - \hat{S}(t, \vec{x}_i))^2$$

However, if the dataset contains samples that are right censored, then it is necessary to adjust the score by weighting the squared distances using the inverse probability of censoring weights method. Let $\hat{G}(t) = P[C > t]$ be the estimator of the conditional survival function of the censoring times calculated using the Kaplan-Meier method, where C is the censoring time.

$$BS(t) = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{\left(0 - \hat{S}(t, \vec{x}_i)\right)^2 \cdot 1_{T_i \le t, \delta_i = 1}}{\hat{G}(T_i^-)} + \frac{\left(1 - \hat{S}(t, \vec{x}_i)\right)^2 \cdot 1_{T_i > t}}{\hat{G}(t)}\right)$$

In terms of benchmarks, a useful model will have a Brier score below 0.25. Indeed, it is easy to see that if $\forall i \in [\![1, N]\!], \hat{S}(t, \vec{x}_i) = 0.5$, then $BS(t) = 0.25$.

The formula for the estimated censoring survival function $\hat{G}(t)$ is:

$$\hat{G}(t) = \prod_{j:Y_{(j)} \le t}\left(1 - \frac{d_j^*}{n_j}\right)$$

Where at each unique observed time point $Y_{(j)}$:

- $Y_{(j)}$ is the j-th ordered observed time.
- d_j^* is the number of **censoring events** that occurred at time $Y_{(j)}$ ($\delta_i^* = 1$).
- n_j is the number of individuals **at risk of being censored** (i.e., not having an event and not being censored previously) just before time $Y_{(j)}$.

In practice, this is an estimated value, $\hat{G}(t)$.

- ***G(t)*** is the estimated probability that an individual is *not* censored before the evaluation time point t specified in the Brier score calculation.
- ***G(T_i)* or *G(Y_i)*** refers to the probability that an individual i is not censored before their own observed time T_i (or Y_i), which is $\min(\text{True Event Time}, \text{Censoring Time})$. When used in the weights, a left-continuous version $\hat{G}(Y_i^-)$ is used to avoid division by zero if an event and a censoring occur at the exact same time.

Meaning and Role in IPCW

The IPCW Brier score at time t uses these terms in its weights to adjust for the bias caused by censoring. The general idea of IPCW is to give more weight to uncensored individuals, particularly those observed for longer periods, to compensate for similar individuals whose event status becomes unknown earlier due to censoring.

The **Integrated Brier Score** (IBS) provides an overall calculation of the model performance at all available times.

$$\text{IBS}(t_{\max}) = \frac{1}{t_{\max}}\int_0^{t_{\max}} BS(t)dt$$

Evaluation Metrics – **Integrated Brier Score (IBS)**

Example: Integrated Brier Score Calculation (Conceptual)

In practice, the IBS is computed using specialized statistical software like the R packages `pec` or `survex`, as manual calculation of the necessary censoring weights ($\hat{G}(t)$) and time integration is complex

1. Setup Data and Model

We have a test set of 4 patients with their observed times (T), event status (Δ), and predicted survival probabilities $S(t)$ at specific time points (e.g., 1 year, 2 years, 3 years)

Patient ID	T (Years)	Δ (Event)	$S(1y)$ Pred	$S(2y)$ Pred	$S(3y)$ Pred
A	0.5	1	0.9	0.8	0.7
B	1.8	1	0.9	0.7	0.5
C	2.5	0	0.8	0.6	0.4
D	3.0	1	0.9	0.7	0.6

We want to evaluate the IBS over the time range $[0, 3 \text{ years}]$.

2. Calculate the Time-Dependent Brier Score (BS) at Specific Time Points

The standard Brier score at a single time point t is the mean squared difference between the observed status (1 if alive at time t, 0 if dead) and the predicted survival probability $S(t)$.

For right-censored data, an Inverse Probability of Censoring Weights (IPCW) method is used to adjust for bias. The formula for the BS at time t for a dataset of N patients is:

$$\text{BS}(t) = \frac{1}{N} \sum_{i=1}^{N} \left(\frac{(0 - \hat{S}(t|\mathbf{x}_i))^2 \cdot \mathbb{I}(T_i \leq t, \Delta_i = 1)}{\hat{G}(T_i)} + \frac{(1 - \hat{S}(t|\mathbf{x}_i))^2 \cdot \mathbb{I}(T_i > t)}{\hat{G}(t)} \right)$$

Where $\hat{G}(t)$ is the estimated probability of being censored after time t (estimated via a Kaplan-Meier model on censoring times of the training data). The terms in the numerator serve to weight observations correctly based on whether they experienced the event, were censored, or survived beyond time t.

Example BS values (calculated by software):

Assuming software calculates the following Brier Scores at the chosen time points:

- $\text{BS}(t = 1 \text{ year}) = 0.10$
- $\text{BS}(t = 2 \text{ years}) = 0.15$
- $\text{BS}(t = 3 \text{ years}) = 0.20$

Evaluation Metrics – **Integrated Brier Score (IBS)**

Example: Integrated Brier Score Calculation (Conceptual)

In practice, the IBS is computed using specialized statistical software like the R packages `pec` or `survex`, as manual calculation of the necessary censoring weights ($\hat{G}(t)$) and time integration is complex

1. Setup Data and Model

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B	1.8	1	0.9	0.7	0.5
C	2.5	0	0.8	0.6	0.4
D	3.0	1	0.9	0.7	0.6

We want to evaluate the IBS over the time range $[0, 3 \text{ years}]$.

3. Integrate the Brier Scores over Time

The IBS is the average of the Brier scores across the entire time range $[t_{min}, t_{max}]$. This integral is numerically approximated (usually using the trapezoidal rule) and normalized by the length of the time interval.

$$\text{IBS} = \frac{1}{t_{\max} - t_{\min}} \int_{t_{\min}}^{t_{\max}} \text{BS}(t)\, dt$$

Using our example values and the trapezoidal rule to approximate the area under the curve from 0 to 3 years:

The area is approximated as the sum of areas of trapezoids formed between the time points.

- Area 1 (0 to 1 year, assuming BS(0) = 0): $0.5 \times (1 - 0) \times (0 + 0.10) = 0.05$
- Area 2 (1 to 2 years): $0.5 \times (2 - 1) \times (0.10 + 0.15) = 0.125$
- Area 3 (2 to 3 years): $0.5 \times (3 - 2) \times (0.15 + 0.20) = 0.175$

Total Area = $0.05 + 0.125 + 0.175 = 0.35$

The time range is 3 years.

$$\text{IBS} = \frac{\text{Total Area}}{\text{Time Range}} = \frac{0.35}{3} \approx 0.117$$

Interpretation

The resulting **IBS of 0.117** is an overall measure of the model's accuracy across the 3-year period.

- The IBS ranges from 0 to 1, with a value of 0 representing a perfect model.
- Lower values indicate better predictive performance and calibration (the model's predicted probabilities match the observed event rates).
- A useful benchmark: a model is generally considered useful if its IBS is below 0.25.

Reference

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